Assignment 2 COAL

25K-0116. Syeda Haram Waqar

BAI-3A

Q.1

AX = 3

26 = 16 + 8 + 2

AX x 26 = AX x 16 + AX x 8 + AX x 2

```
mov  ax, 3

mov  bx, ax
sht  bx, 4      ; BX = 3 x 16 = 48

mov  cx, ax
shl  cx, 3      ; CX = 3 x 8 = 24

Sh ax, 1        ; ax = 3 x 2 = 6

add  ax, bx     ; AX = 6 + 48 = 54
add  ax, cx     ; AX = 54 + 24 = 78
```

AX = 78 final answer

Q.2

Date

Q.2

```
.data
val DWORD 02345ABCDh
.code
main PROC
    mov eax, val
    mov ecx, 4

ShiftRightLoop:
    ror eax, 8
    Loop ShiftRightLoop

    mov eax, val
    mov ecx, 4
ShiftLeftLoop:
    rol eax, 8
    Loop ShiftLeftLoop

    ret.
```

Q.3

```
.data
val dword 02345ABCDh
.code
main PROC
    mov eax, val
    ror eax, 16
    ror eax, 16

    rol eax, 16
    rol eax, 16

    ret.
```

Signature

No.

Date

Q:4

DX = 0 1011 0011 1101 1010

```
.data
BSeconds   BYTE ?
BMinutes   BYTE ?
BHours     BYTE ?

.code
main PROC

; DX = given
;(a)  Extract seconds (bits 0-4)
mov  ax, dx
and  ax, 001Fh
mov  BSeconds, al

; (b) Extract minutes (bit 5-10)
mov  ax, dx
shr  ax, 5
and  ax, 003Fh
mov  BSeconds, al

; (C) Extract hours (bit 11-15)
mov  ax, dx
shr  ax, 11
and  ax, 001Fh
mov  BHours, al
```

Date

Q: 5

```
mov  dx, 9A3Fh      ; 1001 1010 0011 1111
mov  ax, 7F80h      ; 0111 1111 1000 0000
clc                 ; CF=0
add  ax, 90EAh      ; CF=1 ← 1 0001 0000 0110 1010
                          (17 bit)
adc  dx, 1          ; dx+1+CF(1)
                    ; dx → 1001 1010 0100 0001
                    ; CF=0
stc                 ; forcefully CF=1
add  ax, 8120h      ; ax → 1001 0001 1000 1010
                    ; CF=0
adc  dx, 0FFFFh     ; dx = 1 0000 1001 1010 0000
                           (17 bit)
                    ; CF=1
```

Final Answer:

DX = 9A40h

AX = 818Ah

CF = 1

Q: 6

(a) code: if ax=0

```
test ax, ax ;
```

Reason → TEST AX, AX sets ZF=1 if AX is zero

(b)

```
test bx, 1
```

Reason → if '0' ZF=1 otherwise '1' ZF=0

No.

Date

(c) Test DX, DX.

Reason → SF = 0 if (+ve) & SF = 1 if (-ve)

(d) Test DX, 8000h

Reason → ZF = 1 if result = condition True

e) Test BL, BL

check even no of 1's

Q:07

```
mov dx, 5        ; dx → 0000 0000 0000 0101
stc              ; CF = 1
mov ax, 10h      ; ax → 0000 0000 0001 0000
adc dx, ax       ; dx → 0000 0000 0001 0110
```

dx = 0016h

CF = 0

Q:08

```
.data
Val1  QWORD  2040300436204 7A1h
Val2  QWORD  055213O4A2630B2h
result QWORD 0
.code
main PROC
mov cx, 8
mov esi, offset Val1
mov edi, offset Val2
mov ebx, offset result
```

Signature

No.

Date

clc

```
top:
    mov  al, BYTE PTR [esi]    ; AL = A
    SBB  al, BYTE PTR [edi]    ; AL = A - B - CF
    mov  BYTE PTR [ebx], al
    inc  esi
    inc  edi
    inc  ebx

    loop top
```

Q.09

```
.data
    marks  DWORD 75
    grade  BYTE ?

.code
    mov  eax, marks

    cmp  eax, 80        ;
    jge  GradeA

    cmp  eax, 60
    jge  GradeB

    cmp  eax, 40
    jge  GradeC

    mov  grade, 'F'
    jmp  Done
```

Signature

No.

Date

```
Grade A:
    mov  grade, 'A'
    Jmp  Done

Grade B:
    mov  grade, 'B'
    Jmp  Done

Grade C:
    mov  grade, 'C'

Done:
```

Q: 10

EAX = 3
EBX = 10
Stack = Empty

Q: 11

```
.data
num1 dword ?
num2 dword ?
num3 dword ?
num4 dword ?
result dword ?
```

Date

•code

```
move  eax, num1
cmp   eax, num1
jle   checkElseIf

mov   eax, num3
cmp   eax, num4
je    checkElseIf

mov   eax, num1
add   eax, num3
mov   result, eax

CheckElseIf:
mov   eax, num2
cmp   eax, num4
JNE   ElsePart

move  eax, num2
sub   eax, num1
mov   result, eax

ElsePart
mov   result, 0
```

Q:12

```
.data
op1 dword ?
op2 dword ?
x   dword ?
y   dword ?
z   dword ?

.code
 OuterLoop:
mov  eax, op1
cmp  eax, op2
jg   Done

Inner Loop:
mov  eax, op1
cmp  eax, op2
jg   outerLoop

mov  eax, op1
cmp  eax, x
jl   ElsePart

mov  eax, op2
cmp  eax, y
jg   ElsePart

~~move eax, 7~~
add  z, 10
jmp  continueLoop

ElsePart:
sub  z, 10

continue Loop:
dec  op1
jmp  InnerLoop

Done:
```